

Scalability & Future Plan

Date	14 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Voice-Based Concept Understanding Analyzer (VBCUA) can be scaled to support multiple users, larger volumes of audio submissions, and enhanced AI powered assessment capabilities in the future. It also presents the team's roadmap for improving the system beyond its current implementation by introducing advanced features, cloud deployment, multilingual support, and institutional integration, ensuring long-term usability, maintainability, and educational impact.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently evaluates one audio response at a time and supports only uploaded audio files.	Limited scalability for large classrooms and real-time assessments.	High
2	The system compares responses against a single reference answer for each concept.	May not accurately evaluate different valid explanations of the same concept.	High
3	The application is deployed locally using Streamlit without user authentication or database support.	Not suitable for multi-user institutional deployment.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user Streamlit application	Deploy on cloud platforms (AWS, Azure, Google Cloud) to support multiple concurrent users
2	Data Storage	Reference answers and reports stored locally.	Integrate PostgreSQL or MongoDB for centralized data storage.
3	Performance	CPU-based Whisper inference may increase processing time	Use GPU acceleration, model optimization, and asynchronous processing
4	Security	No authentication or access control	Implement secure login, role-based access, and encrypted data storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Real-time microphone recording and support for multiple reference answers	Jan – Jun 2027	Improved user experience and more accurate concept evaluation
Phase 3	Multilingual support, educator dashboard, and question bank integration	Jul – Dec 2027	Support for diverse learners and better performance tracking
Phase 4	Cloud deployment, mobile application, and AI-powered personalized feedback	Jan – Jun 2028	Institution-scale deployment and enhanced learning experience